FINGERPRINT ENROLLMENT CODE

The following code is uploaded to the ESP32 to enrol each voter's fingerprint before the election. The voter ID entered in the Serial Monitor (1–5) maps to Voter IDs V001–V005.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Adafruit_Fingerprint.h>
//   FINGERPRINT ENROLLMENT — ESP32
//   FP  : RX2=GPIO16, TX2=GPIO17
//   LCD : SDA=D21, SCL=D22
//   Open Serial Monitor at 115200 baud
//   Enter ID (1-5) in Serial Monitor to enroll
// ---------- LCD ----------
LiquidCrystal_I2C lcd(0x27, 16, 2);
// ---------- FINGERPRINT (UART2) ----------
HardwareSerial fpSerial(2);
Adafruit_Fingerprint finger(&fpSerial);
// ---------- LCD HELPER ----------
void lcdPrint(String line1, String line2 = "") {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(line1);
  if (line2 != "") {
    lcd.setCursor(0, 1);
    lcd.print(line2);
  }
}
// ---------- ID variable ----------
uint8_t id;
//   READ NUMBER FROM SERIAL MONITOR
uint8_t readnumber(void) {
  uint8_t num = 0;
  while (num == 0) {
    while (!Serial.available());
```

```
    char c = Serial.read();
    if (isdigit(c)) {
      num *= 10;
      num += c - '0';
    }
  }
  return num;
}
//   SETUP
void setup() {
  Serial.begin(115200);
  delay(100);
  // LCD
  Wire.begin(21, 22);
  lcd.init();
  lcd.backlight();
  lcdPrint("FP Enrollment   ", "Starting...     ");
  delay(1000);
  // Fingerprint UART2
  fpSerial.begin(57600, SERIAL_8N1, 16, 17);
  finger.begin(57600);
  Serial.println("\n\nFingerprint Enrollment — ESP32");
  if (finger.verifyPassword()) {
    Serial.println("Fingerprint sensor found!");
    lcdPrint("Sensor Found!   ", "Ready to Enroll ");
  } else {
    Serial.println("Fingerprint sensor NOT found!");
    lcdPrint("Sensor Error!   ", "Check Wiring    ");
    while (1) { delay(1); }
  }
  delay(2000);
  Serial.println("\nEnter ID (1-5) in Serial Monitor to begin enrollment.");
  lcdPrint("Enter ID (1-5)  ", "Serial Monitor  ");
}
```

```
//   ENROLL FINGERPRINT FUNCTION
uint8_t getFingerprintEnroll() {
  int p = -1;
  Serial.print("Waiting for finger to enroll as ID #");
  Serial.println(id);
  lcdPrint("Place Finger    ", "ID: " + String(id));
  // ---- Step 1: Get first image ----
  while (p != FINGERPRINT_OK) {
    p = finger.getImage();
    switch (p) {
      case FINGERPRINT_OK:
        Serial.println("Image taken");
        lcdPrint("Image Taken!    ", "ID: " + String(id));
        break;

      case FINGERPRINT_NOFINGER:
        Serial.print(".");
        // No update on LCD — keep showing "Place Finger"
        break;
      case FINGERPRINT_PACKETRECIEVEERR:
        Serial.println("Communication error");
        lcdPrint("Comm Error!     ", "Try again       ");
        break;
      case FINGERPRINT_IMAGEFAIL:
        Serial.println("Imaging error");
        lcdPrint("Imaging Error!  ", "Try again       ");
        break;
      default:
        Serial.println("Unknown error");
        lcdPrint("Unknown Error!  ", "Try again       ");
        break;
    }
  }
  // ---- Step 2: Convert first image ----
```

```
p = finger.image2Tz(1);
switch (p) {
  case FINGERPRINT_OK:
    Serial.println("Image converted");
    lcdPrint("Image Converted ", "Remove Finger   ");
    break;
  case FINGERPRINT_IMAGEMESS:
    Serial.println("Image too messy");
    lcdPrint("Image Messy!    ", "Try again       ");
    return p;
  case FINGERPRINT_PACKETRECIEVEERR:
    Serial.println("Communication error");
    lcdPrint("Comm Error!     ", "Try again       ");
    return p;
  case FINGERPRINT_FEATUREFAIL:
    Serial.println("Could not find fingerprint features");
    lcdPrint("Feature Error!  ", "Try again       ");
    return p;
  case FINGERPRINT_INVALIDIMAGE:
    Serial.println("Invalid image");
    lcdPrint("Invalid Image!  ", "Try again       ");
    return p;
  default:
    Serial.println("Unknown error");
    lcdPrint("Unknown Error!  ", "Try again       ");
    return p;
}
// ---- Step 3: Remove finger ----
Serial.println("Remove finger");
lcdPrint("Remove Finger   ", "Wait...         ");
delay(2000);
p = 0;
while (p != FINGERPRINT_NOFINGER) {
  p = finger.getImage();
```

```
}
Serial.print("ID: "); Serial.println(id);
// ---- Step 4: Get second image ----
p = -1;
Serial.println("Place same finger again");
lcdPrint("Place Same      ", "Finger Again    ");
while (p != FINGERPRINT_OK) {
  p = finger.getImage();
  switch (p) {
    case FINGERPRINT_OK:
      Serial.println("Image taken");
      lcdPrint("Image Taken!    ", "Processing...   ");
      break;
    case FINGERPRINT_NOFINGER:
      Serial.print(".");
      break;
    case FINGERPRINT_PACKETRECIEVEERR:
      Serial.println("Communication error");
      lcdPrint("Comm Error!     ", "Try again       ");
      break;
    case FINGERPRINT_IMAGEFAIL:
      Serial.println("Imaging error");
      lcdPrint("Imaging Error!  ", "Try again       ");
      break;

    default:
      Serial.println("Unknown error");
      lcdPrint("Unknown Error!  ", "Try again       ");
      break;
  }
}
// ---- Step 5: Convert second image ----
p = finger.image2Tz(2);
switch (p) {
```

```
  case FINGERPRINT_OK:
    Serial.println("Image converted");
    lcdPrint("Image Converted ", "Creating Model  ");
    break;
  case FINGERPRINT_IMAGEMESS:
    Serial.println("Image too messy");
    lcdPrint("Image Messy!    ", "Try again       ");
    return p;
  case FINGERPRINT_PACKETRECIEVEERR:
    Serial.println("Communication error");
    lcdPrint("Comm Error!     ", "Try again       ");
    return p;
  case FINGERPRINT_FEATUREFAIL:
    Serial.println("Could not find features");
    lcdPrint("Feature Error!  ", "Try again       ");
    return p;
  case FINGERPRINT_INVALIDIMAGE:
    Serial.println("Invalid image");
    lcdPrint("Invalid Image!  ", "Try again       ");
    return p;
  default:
    Serial.println("Unknown error");
    lcdPrint("Unknown Error!  ", "Try again       ");
    return p;
}
// ---- Step 6: Create model ----
Serial.print("Creating model for ID #"); Serial.println(id);
lcdPrint("Creating Model  ", "Please wait...  ");
p = finger.createModel();

if (p == FINGERPRINT_OK) {
  Serial.println("Fingerprints matched!");
  lcdPrint("Prints Matched! ", "Storing...      ");
}
```

```
else if (p == FINGERPRINT_PACKETRECIEVEERR) {
    Serial.println("Communication error");
    lcdPrint("Comm Error!     ", "Try again       ");
    return p;
  }
 else if (p == FINGERPRINT_ENROLLMISMATCH) {
    Serial.println("Fingerprints did not match — try again");
    lcdPrint("Mismatch!       ", "Try again       ");
    return p;
  }
else {
    Serial.println("Unknown error");
    lcdPrint("Unknown Error!  ", "Try again       ");
    return p;
  }
  // ---- Step 7: Store model ----
  Serial.print("Storing ID #"); Serial.println(id);
  p = finger.storeModel(id);
  if (p == FINGERPRINT_OK)
{
    Serial.println("Stored successfully!\n");
    lcdPrint("Stored! ID: " + String(id), "Enrollment Done!");
    delay(3000);
  }
else if
(p == FINGERPRINT_PACKETRECIEVEERR) {
    Serial.println("Communication error");
    lcdPrint("Comm Error!     ", "Try again       ");
    return p;
  }
else if
(p == FINGERPRINT_BADLOCATION) {
    Serial.println("Could not store at that location");
    lcdPrint("Bad Location!   ", "Try again       ");
```

```
    return p;
  }
else if
(p == FINGERPRINT_FLASHERR) {
    Serial.println("Flash write error");
    lcdPrint("Flash Error!    ", "Try again       ");
    return p;
  }
else
 {
    Serial.println("Unknown error");
    lcdPrint("Unknown Error!  ", "Try again       ");
    return p;
  }
  return true;
}
//   MAIN LOOP
void loop() {

  Serial.println("\nReady to enroll!");
  Serial.print("Enter ID (1-5): ");
  lcdPrint("Enter ID (1-5)  ", "Serial Monitor  ");
  id = readnumber();
  // Validate ID
  if (id == 0 || id > 5) {
    Serial.println("Invalid! Enter 1-5 only.");
    lcdPrint("Invalid ID!     ", "Enter 1 to 5    ");
    delay(2000);
    return;
  }
  Serial.print("Enrolling ID #");
  Serial.println(id);
  lcdPrint("Enrolling...    ", "ID: " + String(id));
  delay(1000);
```

```
  // Keep trying until enrollment succeeds
  while (!getFingerprintEnroll());
  Serial.println("Next enrollment ready.");
  lcdPrint("Done! ID: " + String(id), "Next: New ID    ");
  delay(2000);
}
```